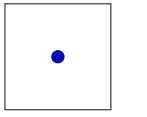
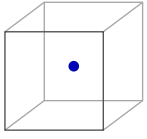
● advected unknown of the cell □ centre value read from a neighbour cell ○ no unknown: value from another cell

OWN: the points of a cell that carry advected unknowns (top: 2D, bottom: 3D)

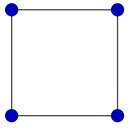
C

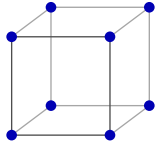




centre

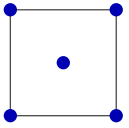
SC

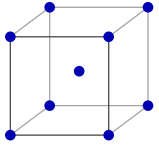




vertices

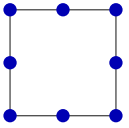
BCC

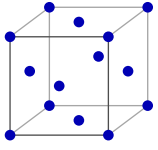




centre + vertices

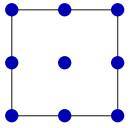
FCC

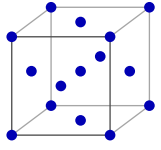




vertices + face centres

BFC

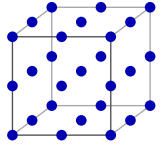




centre + vertices + face centres

T

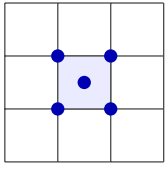
in 2D equal to BFC



BFC + edge midpoints

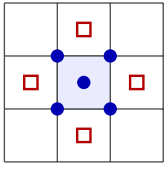
NEIGHBOUR: centre values read from other cells

BCC



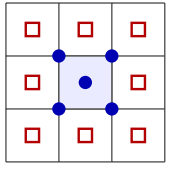
no suffix: own points only

BCC-Cf



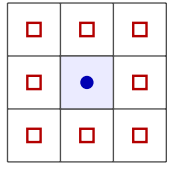
`-Cf`: the face neighbours

BCC-Cp



`-Cp`: the point neighbours

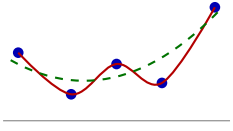
C-Cp



the original scheme

SPACE

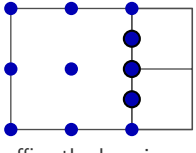
`/I`: returns every datum



`/L2`: least-squares quadratic

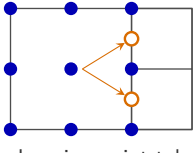
TRANSITION: the treatment of a 2:1 refinement transition

BFC/I



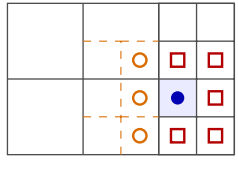
no suffix: the hanging points (black rim) are unknowns

BFC/I+c



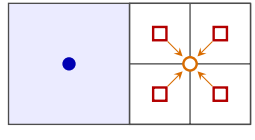
+c: a hanging point takes the interpolant of the coarse cell

C-Cp/L2+g



+g, fine cell: ghost values in a coarse cell from its reconstruction

C-Cp/L2+g



+g, coarse cell: ghost value = mean of the fine values